CHRISTOPHER KOPIWODA

Kenilworth, IL 60043 | (847) 840-2489 | hello@kopiwoda.me
linkedin.com/in/christopher-kopiwoda | christopherkopiwoda.com

EDUCATION

Michigan State University, College of Engineering East Lansing, MI
B.S. Mechanical Engineering; Minor in Materials Science Engineering | M.S. Mechanical Engineering Planned Expected May 2027
GPA: 3.82/4.00 | Honors College | Tau Beta Pi | Pi Tau Sigma

ENGINEERING EXPERIENCE

Grayhill La Grange, IL
Mechanical Engineering Intern May 2026 – Aug. 2026

- Redesigned a **continuity tester in Creo** capable of checking approximately **56 electrical components in ~15 seconds**; iterated with 3D-printed prototypes and created manufacturing drawings for final fabrication.
- Built and soldered the complete electrical hardware for a **48-channel Chatterbox vibration-test monitoring system**; completed the in-house unit for approximately **$80 in parts**, avoiding another approximately **$20,000 commercial Chatterbox purchase**.
- Programmed the Chatterbox to detect switch-state changes exceeding **10 µs**, developed a **Python GUI** interfacing with **Arduino firmware**, and validated timing performance with an **oscilloscope and signal generator** before internal deployment.
- Developed **Python static-analysis scripts** to calculate joystick-holder tipping force across varying height, width, and weight parameters, using the results to select holder dimensions.

Michigan State University — Zevalkink Research Group East Lansing, MI
Research Assistant | NASA Michigan Space Grant Undergraduate Research Fellow Aug. 2025 – Present

- Awarded a **NASA Michigan Space Grant Consortium Undergraduate Research Fellowship** to continue thermoelectric materials research during Fall 2026.
- Develop an ampule-based optical traveling zone process for Bi_2Se_3 crystal growth using an Optical Floating Zone furnace; prepare high-purity materials, vacuum-seal quartz ampules, and perform XRD characterization.
- Calibrated furnace behavior and established approximately **40% optical power** as a reproducible Bi_2Se_3 melting threshold, creating a repeatable baseline for subsequent growth experiments.

LEADERSHIP & TEACHING

Michigan State University — Math Learning Center East Lansing, MI
Supervisor; previously Undergraduate Learning Assistant Aug. 2024 – May 2026

- Selected for advancement from Undergraduate Learning Assistant to **Math Learning Center Supervisor**, overseeing approximately **20 tutors** across two semesters.
- Previously taught **4 calculus recitation sections** of approximately **20–30 students each**, providing in-class and out-of-class instruction and support.

SELECTED ENGINEERING PROJECTS

Motor-Powered Rope Traverser Michigan State University
Mechanical Design & Manufacturing II | SolidWorks, Abaqus, MATLAB Spring 2026

- Designed and built a shoebox-sized robot using synchronized **Grashof four-bar linkages** to traverse a 10-meter suspended rope; created and iterated much of the CAD and fabricated components using **machining and 3D printing**.
- Performed complete **Abaqus FEA of the drive shaft**, validated results against analytical calculations, and helped execute a late-stage redesign involving new CAD, machining, reprinting, assembly, and testing.

Yolkey — MSU Designathon Michigan State University
Mechanical Design Lead | Siemens NX, Mechanism Design March 2025

- Led CAD and mechanical-system design of a battery-free assistive key-turning device using a **Scotch-yoke mechanism and multi-stage gear reduction** to increase available turning torque.
- Won the **Accessibility & Inclusivity Design Track** and **Organizer's Favorite** awards.

Ghomeo — SpartaHack X Michigan State University
Frontend Development | React Native, JavaScript, Firebase February 2025

- Built the frontend and dynamic garden-layout system for a cross-platform garden-planning app; **won the Sustainability Track** at SpartaHack X among 498 participants from 38 schools.

Self-Hosted AI & Automation Server Personal Project
Ubuntu Server | Ollama, Qwen3.5 4B, n8n, Docker, JMAP, REST APIs 2026 – Present

- Deployed a **self-hosted Qwen3.5 4B LLM with Ollama** on a headless Ubuntu server for private local inference alongside containerized services.
- Built an **n8n email-categorization workflow** using **Fastmail JMAP, REST APIs, and structured local-LLM outputs** to automatically classify and organize incoming email.

TECHNICAL SKILLS

CAD & Analysis: SolidWorks, Creo, Siemens NX, Fusion 360, Abaqus, FEA, MATLAB
Manufacturing & Testing: Machining, Milling, Lathe, 3D Printing, Rapid Prototyping, Soldering, Oscilloscope, Signal Generator, XRD
Programming, AI & Automation: Python, C++, Java, LLM Applications, Agentic Workflows, n8n, APIs, Docker, Git/GitHub, Linux